

Qudit Pauli and Finite-Heisenberg Sigma–MGFs

Proof, matrix construction, and direct checks

Computational verification note

17 July 2026

Abstract

For odd characteristic we prove the two-spectrum formula for the defining Schrödinger representation and check it by constructing every matrix in $P_{3,1}$, $P_{5,1}$, $P_{9,1}$, and $H_3(\mathbb{F}_9)$. The tests evaluate the target invariant dimension directly, not from the spectral classification used in the proof. All 17 representative coefficient comparisons pass.

1 Statement

Let $q = p^f$ with p odd, $V = \mathbb{C}[\mathbb{F}_q^n]$, and $N = q^n$. Fix a nontrivial additive character $\psi : \mathbb{F}_q \rightarrow \mu_p$. With

$$X(a)e_x = e_{x+a}, \quad Z(b)e_x = \psi(b \cdot x)e_x, \quad W(a, b) = \psi\left(\frac{1}{2}a \cdot b\right) X(a)Z(b),$$

put $P_{q,n} = \{\zeta W(a, b) : \zeta \in \mu_p\}$. For a partition $\tau = (\tau_1, \dots, \tau_\ell)$ define

$$A_\tau = \prod_i \binom{N + \tau_i - 1}{\tau_i}, \quad B_\tau = \begin{cases} \prod_i \binom{N/p + \tau_i/p - 1}{\tau_i/p}, & p \mid \tau_i \ \forall i, \\ 0, & \text{otherwise.} \end{cases}$$

Then the claimed generalized Molien coefficient is

$$[m_\tau] \mathcal{S}_{P_{q,n}} = \begin{cases} 0, & p \nmid |\tau|, \\ q^{-2n} A_\tau + (1 - q^{-2n}) B_\tau, & p \mid |\tau|. \end{cases} \quad (1)$$

Equivalently,

$$\mathcal{S}_{P_{q,n}}(\mathbf{t}) = \frac{q^{-2n}}{p} \sum_{\zeta \in \mu_p} \prod_j (1 - \zeta t_j)^{-N} + (1 - q^{-2n}) \prod_j (1 - t_j^p)^{-N/p}. \quad (2)$$

2 Proof

There are p central matrices ζI_N , a fraction q^{-2n} of the group. Their average is the first term of (2). Now take $(a, b) \neq (0, 0)$. For $1 \leq k < p$, $W(a, b)^k$ has trace zero: if $ka \neq 0$ it is a translation with zero diagonal; otherwise its diagonal trace is a nontrivial additive-character sum. Also $W(a, b)^p = I$. Fourier inversion of the multiplicities of the p th roots of unity therefore gives multiplicity N/p for every root. Hence

$$\det(I - tW(a, b)) = (1 - t^p)^{N/p}.$$

A scalar in μ_p only permutes this spectrum, proving (2). Expanding the central factors gives A_τ and forces $p \mid |\tau|$; expanding the noncentral factors gives B_τ . This proves (1).

For the finite Heisenberg group $H_{2n+1}(\mathbb{F}_q)$, the Schrödinger matrix is $\rho_\psi(a, b, z) = \psi(z)W(a, b)$. When $f > 1$, $\ker \psi$ has size q/p , so each matrix in $P_{q,n}$ has exactly q/p preimages. Uniform averages over the abstract group and its image consequently agree.

3 Independent verification method

For every constructed matrix g , the code computes its eigenvalues $\lambda_1, \dots, \lambda_N$ numerically and obtains $h_d(\lambda)$ from

$$\prod_{r=1}^N (1 - \lambda_r t)^{-1} = \sum_{d \geq 0} h_d(\lambda) t^d.$$

It then evaluates the target directly as

$$D_\tau = \frac{1}{|G|} \sum_{g \in G} \prod_i h_{\tau_i}(\text{spec } g).$$

This is the character of $\bigotimes_i \text{Sym}^{\tau_i} V$ averaged over G , hence the rank of its Reynolds projector. Only after computing D_τ does the program evaluate (1).

The $q = 9$ construction uses $\mathbb{F}_9 = \mathbb{F}_3[u]/(u^2 + 1)$ and $\psi(x) = \exp(2\pi i \text{Tr}_{\mathbb{F}_9/\mathbb{F}_3}(x)/3)$. Thus the test includes a genuine prime-power field. The $H_3(\mathbb{F}_9)$ run enumerates all 729 abstract triples, retaining the threefold central kernel rather than silently replacing the group by its image.

4 Representative results

group	dim V	matrices averaged	τ	direct = formula
$P_{3,1}$	3	27	(3)	2
$P_{3,1}$	3	27	(3, 3)	12
$P_{5,1}$	5	125	(4, 1)	14
$P_{5,1}$	5	125	(5, 5)	636
$P_{9,1}$	9	243	(3)	5
$P_{9,1}$	9	243	(3, 3)	345
$H_3(\mathbb{F}_9)$	9	729	(2, 1)	5
$H_3(\mathbb{F}_9)$	9	729	(3, 3)	345

All 17 built-in cases pass to tolerance 2×10^{-8} and round to nonnegative integers. Characteristic two is intentionally excluded here: the displayed half-phase normalization is an odd-characteristic statement; the phase-extended qubit lift is handled in its own verification.

5 Reproduction

Run

```
.venv/bin/python for_this_guy/quantum_matrix_groups/
  pauli_heisenberg/verification.py
.venv/bin/marimo edit for_this_guy/quantum_matrix_groups/
  pauli_heisenberg/notebook.py
```

The line breaks after the directory slash are for display; join each command onto one shell line.